

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

04-10-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

PAPER-2

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **25 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 5 **Numerical Value Type** questions
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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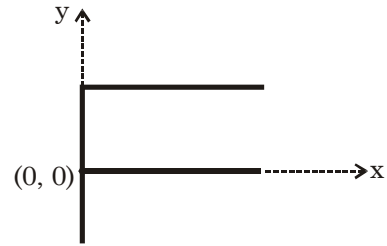
ALLEN
PART 1 - PHYSICS

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

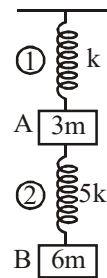
1. Brakes are applied at $t = 0$ in a car travelling with a speed of 72 km/hr which produces a retardation of 5 m/s^2 . Displacement of car in 0 to 6 seconds is
- (A) 30 m
(B) 20 m
(C) 40 m
(D) 50 m

2. Letter F is formed from three identical rods of length ℓ . Find coordinates of its centre of mass.



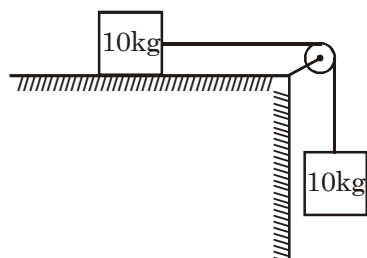
- (A) $\left(\frac{\ell}{5}, \frac{\ell}{5}\right)$ (B) $\left(\frac{\ell}{3}, \frac{\ell}{6}\right)$
 (C) $\left(\frac{\ell}{6}, \frac{\ell}{3}\right)$ (D) $\left(\frac{\ell}{2}, \frac{\ell}{5}\right)$

3. If after equilibrium is achieved the spring (2) is cut than find the value of acceleration of block 'A' just after cutting of springs :-



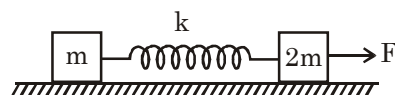
- (A) g (upward) (B) $6g$ (upward)
 (C) $2g$ (upward) (D) g down ward

4. String and pulley are ideal and coefficient of friction between all surfaces is 0.2 then calculate the magnitude of acceleration of blocks:

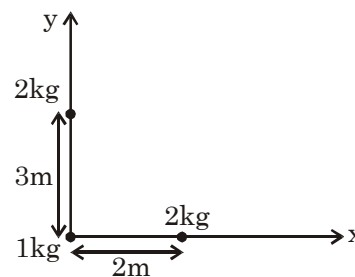


- (A) 2 m/s^2 (B) 4 m/s^2
 (C) 6 m/s^2 (D) 8 m/s^2
5. A ball is projected from a point at two different angles with same speed u and land at same point in both the cases
- (A) The difference between two angles of projection is 90° .
 (B) The maximum height attained by the ball in both the cases is equal.
 (C) The sum of maximum heights for the two cases is $u^2/(2g)$.
 (D) The maximum height attained by the ball in one case must be twice of the maximum height attained by the ball in second case.

6. Given system lies on smooth horizontal surface. Acceleration of center of mass of given system (two blocks + spring) is :

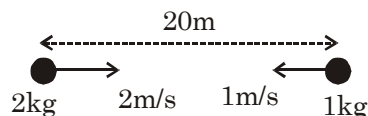


- (A) $\frac{F}{m}$ (B) $\frac{F}{2m}$
 (C) $\frac{F}{3m}$ (D) $\frac{F}{4m}$
7. Position of three particles is shown in diagram. If x and y coordinates of centre of mass of a system of three particles is x_{cm} & y_{cm} then :

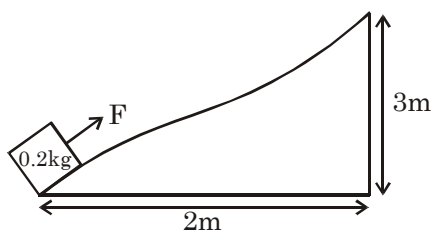


- (A) $x_{\text{cm}} = \frac{2}{5}$ (B) $x_{\text{cm}} = \frac{3}{5}$
 (C) $y_{\text{cm}} = \frac{4}{5}$ (D) $y_{\text{cm}} = \frac{6}{5}$

8. Position & velocity of two particles moving along x-axis is shown in diagram at $t = 0$. What is the displacement of centre of mass of the given system of two particles in 5sec?



- (A) 2m (B) 5m
(C) 4m (D) Zero
9. A body of mass 0.2 kg was slowly pushed up the hill shown in figure by a force F which at each point was directed along a tangent to the trajectory. Find the work performed by this force, if the height of the hill is 3m and length of the base is 2m. Coefficient of kinetic friction (μ_k) = 0.5 (given $g = 10\text{m/s}^2$)



- (A) 4J (B) 2J
(C) 8J (D) 10J

10. If potential energy in space is given as $U = x^2 + 2xy - y^3$, find out value of force at (3, 4, 5).

- (A) $14\hat{i} + 21\hat{j}$
(B) $-14\hat{i} + 42\hat{j}$
(C) $21\hat{i} + 14\hat{j}$
(D) $7\hat{i} + 14\hat{j}$

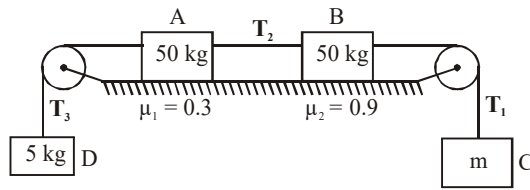
11. A 80 kg man is riding on a 40 kg cart travelling at a speed of 2.5m/s on a frictionless surface. He jumps off the cart such that his velocity just after jump is zero with respect to ground. Select the **CORRECT** statement :

- (A) The work done by the man on the system during the jump is 750 J
(B) Work done by the man on the cart during the jump is 500J
(C) Work done by the man on cart is 750 J
(D) Velocity of the cart after the jump is 5m/s

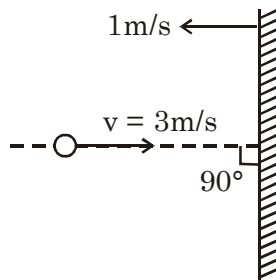
12. A ferry boat is sailing at 12 km/hr 30°W of N with respect to a river that is flowing at 6.0 km/hr E. As observed from the shore, the ferry boat is sailing

- (A) 30°E of N
(B) due N
(C) 30°W of N
(D) 45°E of N

13. A block pulley arrangement is shown in the figure. Find the maximum value of m so that the system will remain at rest.

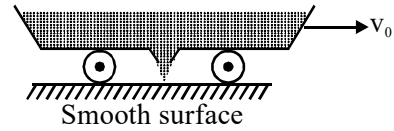


- (A) 50 kg (B) 55 kg (C) 65 kg (D) 75 kg
14. A ball is dropped from height h on stationary horizontal floor. After collision it reaches to maximum height $\frac{h}{4}$ then coefficient of restitution between ball and floor is :-
- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) 1
15. A ball moves with speed 3 m/s towards a heavy wall moving towards the ball with speed 1 m/s. The collision of the ball with the wall is inelastic ($e = 1/2$). Find the speed of ball after collision with the wall.

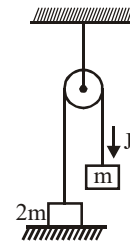


- (A) 2 m/s (B) 3 m/s
(C) 4 m/s (D) 5 m/s

16. A trolley filled with sand moves on a smooth horizontal surface with a velocity v_0 . A small hole is made at the base of it from which sand is leaking out at constant rate. As the sand leaks out then which of the following is **INCORRECT** :-



- (A) The velocity of the trolley increases
(B) The velocity of the trolley remains constant
(C) The momentum of the trolley is conserved
(D) The momentum of the total system (trolley with sand + leaked out sand) is conserved in horizontal direction
17. In the given figure an impulse J is given to the block of mass m in the downward direction. As a result of the impulse :-



- (A) Both the blocks start moving with $\frac{J}{3m}$
(B) Both the blocks start moving with $\frac{J}{m}$ in opposite direction
(C) The centre of mass of the system is moving downwards
(D) The centre of mass of the system is not moving

18. A particle with mass m and initial velocity v_0 is subjected to a velocity dependent force of the form bv^n . Let stopping time t of particle depend on m , v_0 and b as $t \propto m^\alpha b^\beta v_0^\gamma$. Powers α , β and γ will be
- (A) $\alpha = 1, \beta = -1, \gamma = 1 - n$
 (B) $\alpha = 2 - n, \beta = -1, \gamma = 2$
 (C) $\alpha = 1, \beta = 1, \gamma = 1 - n$
 (D) $\alpha = -1, \beta = -1, \gamma = 1 - n$
19. Which of the following statement is **CORRECT** :
- (A) Internal forces can change the linear momentum of a system.
 (B) Internal forces can change the kinetic energy of a system.
 (C) Work done by external forces on a system will always equal to change in kinetic energy of system.
 (D) Work done by internal forces on a system will always equal to zero.
20. Consider two carts, of masses m and $2m$, at rest on an air track. If you push both the carts for 3s exerting equal force on each, the momentum of the light cart is :-
- (A) four times the momentum of the heavy cart
 (B) twice the momentum of the heavy cart
 (C) equal to the momentum of the heavy cart
 (D) one-half the momentum of the heavy cart

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

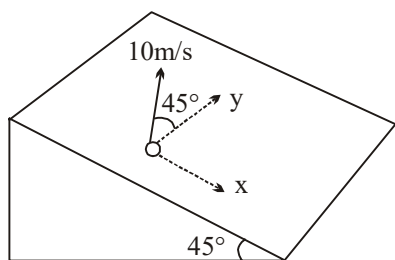
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
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9	9	9	9	9	9	9	9	9	9

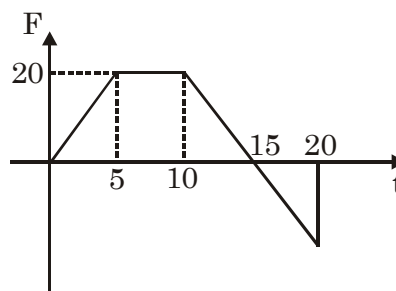
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. The small marble is projected with a velocity of 10 m/s in a direction 45° from the horizontal y -direction on the smooth inclined plane. Calculate the magnitude v of its velocity (in m/s) after 2 seconds. Take $g = 10 \text{ m/s}^2$.



2. An object of mass 4 kg falls from rest through a vertical distance of 10 m and reaches ground with a speed of 14 m/s . How much work is done against air friction in joules? [$g = 10 \text{ m/s}^2$]
3. A car is travelling on the circular road of radius 300 m . At some instant its speed is 30 m/s and is increasing at a rate of 4 m/s^2 . What is its acceleration (in m/s^2)?

4. A toy car can deliver a constant power of 20 W . The resistive force on the car is αv , where v is velocity of car in m/s . If maximum velocity of car is 2 m/s , the value of α is.
5. $F-t$, graph of a particle starts from rest is shown. At what time (in sec), linear momentum of the particle is maximum?



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PART 2 - CHEMISTRY

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The first ionization energy will be maximum for :-
 (A) Helium (B) Barium
 (C) Nitrogen (D) Berellium
2. Which of the following allotropic form is insoluble in carbondisulphide.
 (A) Rhombic sulphur
 (B) Monoclinic sulphur
 (C) White phosphorus
 (D) Red phosphorus
3. Which of the following oxy acid does not have bridging oxy linkage?
 (A) Pyrosulphurous acid
 (B) Pyrophosphoric acid
 (C) Pyrosilicic acid
 (D) Dichromic acid

4. Which of the following statements is **INCORRECT** regarding N_2O_5 ?
 (A) The hybridisation of N in N_2O_5 is sp^2 .
 (B) The total number of lone pair electrons present on N is 2
 (C) It has total 12 lone pair of electrons present
 (D) It has dative bond(s)
5. Which of the following set of quantum numbers represent maximum number of electrons in Ni (Nickel)
 (A) $n = 2, \ell = 1, m = +1$
 (B) $n = 3, \ell = 2, m = 0$
 (C) $n = 3, m = 0, s = +\frac{1}{2}$
 (D) $n = 4, m = 0, s = -\frac{1}{2}$
6. Which of the following species does **NOT** exist.
 (A) SH_2
 (B) PI_6^-
 (C) SeF_6
 (D) XeF_5^-
7. A gas at a pressure of 5.0 atm is heated from $0^\circ C$ to $546^\circ C$ and simultaneously compressed to one-third of its orginal volume. Hence final pressure is
 (A) 10 atm
 (B) 30 atm
 (C) 45 atm
 (D) 5 atm

8. Density of an ideal gas at 2 atm and 600 K

is 2gm/litre. $\left(\text{Given } R = \frac{1}{12} \frac{\text{atm} \times \ell}{\text{mol} \times \text{K}} \right)$

Incorrect options is -

- (A) Vapour density of this gas is 25
 (B) Molar mass of this gas is 50 gm /mol
 (C) Relative density of this gas with respect to Ne (Atomic weight = 20) is 2.5
 (D) Molar mass this gas is 25 gm /mol
9. A regular cube of metal measures exactly 10 cm on an edge and has density 8g/cc. If the cube contains 6×10^{25} atoms of the metal, determine atomic weight of metal ?
 (take $N_A = 6 \times 10^{23}$)
 (A) 40 (B) 60
 (C) 80 (D) 100
10. Number of neutrons in 5.5 gm T_2O (T is ${}_1H^3$) are -
 (A) $0.25 N_A$ (B) $2.5 N_A$
 (C) $3N_A$ (D) $4 \times N_A$
11. 500 ml 3M - H_2O_2 solution is left open due to which some H_2O_2 decomposes and mass of O_2 gas evolved is 8gm at 1atm and $0^\circ C$. Find new molarity of left H_2O_2 solution. [Volume of solution remains constant]
 (Assume : P = 1 atm, T = $0^\circ C$ everywhere)
 (A) 1M (B) 2M
 (C) 0.5 M (D) 1.5 M

12. Calculate the mass of anhydrous oxalic acid ($H_2C_2O_4$), which can be oxidised to CO_2 (g) by 100 ml of an MnO_4^- solution, 10 ml of which is capable of oxidising 50 ml of 1N I^- to I_2 in acidic medium

(A) 45 gm (B) 22.5 gm
 (C) 30 gm (D) 12.25 gm

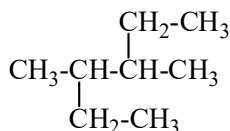
13. What is the molarity of a 4.9% H_2SO_4 solution of density 0.98 gm/ml ? (S = 32)

(A) 0.98 M (B) 0.49 M
 (C) 1M (D) 0.25 M

14. A compound $\begin{array}{c} G \\ | \\ G-C-G \\ | \\ G \end{array}$ contain σ/π ratio, 1 : 1 if G is -

(A) CHO (B) CN
 (C) COOH (D) $C \equiv CH$

15. IUPAC name of compound



(A) 2, 3-diethyl butane
 (B) 2-ethyl-3-methyl pentane
 (C) 3-methyl-2-ethyl pentane
 (D) 3,4-dimethyl hexane

16. Which of the following represent same functional group.

(A) $Me-CH_2-NH_2$ $Me-NH-Me$
 (B) $Me-CH_2-OH$ $Me-O-Me$
 (C) $Me-CH_2-OH$ Me_2CH-OH

(D) $Me-\overset{\overset{O}{\parallel}}{C}-NH_2$ $Me-\overset{\overset{O}{\parallel}}{C}-NH-Me$

- (A) 
- (B) 
- (C) 
- (D) 

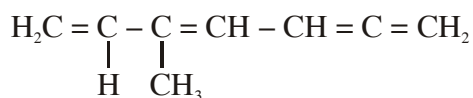
- CN1C(C(=O)CC(C)=O)C2C(C1)C(C(=O)O)C(C2)C(C(=O)O)C3C(C(=O)O)C(C(=O)O)C3

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Figure 1 shows two 9x6 grids of numbers 0-9. The left grid is titled with a plus sign (+) and the right grid is titled with a minus sign (-). Both grids contain a sequence of numbers from 0 to 9 in a 3x3 grid pattern, with some numbers replaced by black squares.

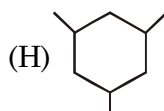
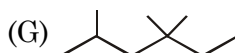
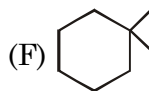
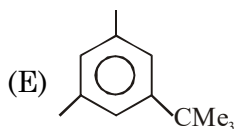
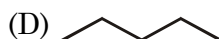
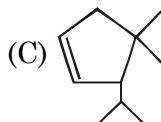
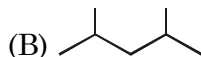
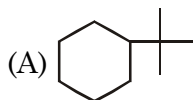
- Nurture Course/Phase-S, I(A), I, II, III, IV(A) & IV/Paper-2**

1. Consider given hydrocarbon



Find number of bond(s) which is/are formed by $\text{sp}^2\text{-sp}^2$ overlapping

2. How many of the following have all type of carbon atoms i.e. (1° , 2° , 3° & 4° C-atoms)



3. 50% pure sample of CaCO_3 heated with 80% yield to produce 0.448 litre CO_2 at 1 atm and 273 K temperature. Calculate mass of CaCO_3 sample in gm.

4. A hydrocarbon C_xH_y on complete combustion gives $\text{CO}_2(\text{g})$ & $\text{H}_2\text{O}(\text{l})$. If volume of $\text{O}_2(\text{g})$ required for complete combustion of hydrocarbon is double the volume of $\text{CO}_2(\text{g})$

produced, then find the value of $\left(\frac{y}{x}\right)$.

5. The bond length in HCl molecule is 0.127 nm. The observed dipole moment of HCl is 1.08 D. The percentage ionic character in HCl molecule is_____.

ALLEN
PART 3 - MATHEMATICS

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
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Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The sum $\sum_{r=1}^{50} (r)$ is-
 (A) 2524 (B) 2525 (C) 1255 (D) 1275
2. ABC is a variable triangle such that A(1, 2) and B and C lie on the line $y = x + \lambda$ ($\lambda \in \mathbb{R}$). Then the locus of the orthocentre of $\triangle ABC$ is
 (A) $x + y = 0$ (B) $x - y = 0$
 (C) $x^2 + y^2 = 4$ (D) $x + y = 3$
3. In $\triangle ABC$, $\tan A = 3$, $\tan B = \frac{2}{3}$, $c = \sqrt{10}$, then circum radius of Δ is
 (A) $11\sqrt{\frac{13}{10}}$ (B) $\frac{10\sqrt{13}}{11}$ (C) $\frac{\sqrt{130}}{11}$ (D) $\frac{5\sqrt{13}}{11}$
4. In $\triangle ABC$ with usual notation, $\frac{c - b \cos A}{a}$ is equal to
 (A) $\cos A$ (B) $\cos B$ (C) $\cos C$ (D) $\sin B$

5. If $\tan \alpha = \frac{5}{6}$ and $\tan \beta = \frac{1}{11}$, α and $\beta \in \left(0, \frac{\pi}{4}\right)$. Then the value of $\cos[\pi - (\alpha + \beta)]$ is
 (A) 1 (B) -1 (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$
6. If A(h,k) divides the line segment joining (2,3) and (-1,2) internally in the ratio 3 : 4, then the value of h + k is
 (A) $\frac{3}{7}$ (B) $\frac{20}{7}$ (C) $\frac{23}{7}$ (D) $\frac{25}{7}$
7. If $1 + \sqrt{2}$ is a roots of the equation $x^2 + bx + c = 0$, where $b, c \in \mathbb{Q}$. Then value of $|b - c|$ is
 (A) 0 (B) 1 (C) 2 (D) 3
8. One diagonal of a square is $3x - 4y + 8 = 0$ and one vertex is (-1, 1) then the area of square is
 (A) $\frac{1}{50}$ sq. units (B) $\frac{1}{25}$ sq. units
 (C) $\frac{3}{50}$ sq. units (D) $\frac{2}{25}$ sq. units
9. Value of $\sin\left(\frac{7\pi}{12}\right) + \cos\left(\frac{7\pi}{12}\right)$ is equal to
 (A) $\sin\left(\frac{\pi}{3}\right)$ (B) $\cos\left(\frac{\pi}{4}\right)$
 (C) $\sin\left(\frac{\pi}{2}\right)$ (D) $\cos\left(\frac{\pi}{3}\right)$

10. Sum of integral value(s) of x satisfying the inequality $\frac{x^2 - 3x - 10}{x^2 + x + 1} \leq 0$, is
(A) 9 (B) 10 (C) 12 (D) 15
11. Value of $\sin^6 x + \cos^6 x + \sin^4 x + \cos^4 x + 3 + 5\sin^2 x \cos^2 x$ is
(A) Prime number
(B) composite number
(C) Negative integer
(D) even number
12. If $D_1 = \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$, $D_2 = \begin{vmatrix} 1 & bc & a \\ 1 & ca & b \\ 1 & ab & c \end{vmatrix}$, then
(A) $D_1 + D_2 = 0$ (B) $D_1 - D_2 = 0$
(C) $D_1 - 2D_2 = 0$ (D) $D_1 + 2D_2 = 0$
13. The value of λ for which the point $(\lambda, 2)$ is interior point of $\triangle ABC$ formed by $x + y = 4$, $3x - 7y = 8$ and $4x - y = 31$ is
(A) $\lambda \in \left(\frac{22}{3}, \frac{33}{4}\right)$ (B) $\lambda \in \left(\frac{21}{3}, \frac{34}{3}\right)$
(C) $\lambda \in \left(\frac{20}{3}, \frac{31}{4}\right)$ (D) $\lambda \in \left(\frac{21}{4}, \frac{33}{5}\right)$
14. The system of equation
 $2x - y + z = 0$
 $x - 2y + z = 0$
 $mx - y + 2z = 0$ has infinite number of non trivial solution for
(A) $m = 2$ (B) $m = -5$
(C) $m = 5$ (D) $m = -2$
15. The image of the point $(4, -13)$ with respect to the line mirror $5x + y + 6 = 0$ is
(A) $(1, 14)$ (B) $(-1, -14)$
(C) $(1, -14)$ (D) $(-1, 14)$
16. Minimum value of the sum of the numbers $a^{-5}, a^{-4}, 2a^{-3}, 6, a^8$ and a^7 is
(A) 5 (B) 6 (C) 11 (D) 12
17. If the point $(t_1, 2at_1 + at_1^3)$, $(t_2, 2at_2 + at_2^3)$ and $(t_3, 2at_3 + at_3^3)$ are collinear where t_1, t_2, t_3 are distinct then the value of $t_1 + t_2 + t_3$ is
(A) 1 (B) 2 (C) 0 (D) $\frac{1}{2}$
18. Sum of infinite terms of series $S = \frac{1}{2.4.6} + \frac{1}{4.6.8} + \frac{1}{6.8.10} + \dots \infty$ is
(A) $\frac{1}{8}$ (B) $\frac{1}{16}$ (C) $\frac{1}{32}$ (D) $\frac{1}{64}$
19. In $\triangle ABC$ with usual notations if $b = 8$, $c = 12$, $\frac{BD}{DC} = \frac{3}{2}$ and $\angle BAD = 30^\circ$ then length of AD is
(A) $\frac{24}{5\sqrt{3}}$ (B) $\frac{24}{5}$ (C) $\frac{24\sqrt{3}}{5}$ (D) $\frac{12}{5}$
20. Let α be the solution of $||4 \sin x| - 1| = 1$ in $\left(0, \frac{\pi}{2}\right)$. If the shadow of a vertical pole is $\frac{1}{\sqrt{3}}$ of its height, then the altitude of the sun is
(A) $\frac{\alpha}{3}$ (B) $\frac{\alpha}{2}$ (C) α (D) 2α

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

					+	●						
●	●	○	○	○	○	○	●	●	●	○	○	○
①	①	①	①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②	②	●	②
③	③	③	③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	●	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Product of the roots of $\begin{vmatrix} 1+x & 1 & 1 \\ 3 & 1 & 2 \\ 3 & x & 3 \end{vmatrix} = 0$ is
2. If $5a + 5b + 20c = t$, then the value of t for which the line $ax + by + c - 1 = 0$ always passes through a fixed point is
3. If $S = 1 + \frac{2}{5} + \frac{3}{5^2} + \frac{4}{5^3} + \dots \infty$, then value of S is
4. If $(2)^{\log_2 x^2} - (3)^{\log_3(x)} - 6 = 0$, then sum, of all possible value(s) of x is
5. The sides of a triangle ABC lie on the lines $3x + 4y = 0$, $4x + 3y = 0$ and $x = 3$. Let (h, k) be the centre of circle inscribed in ΔABC . Then the value of $(h + k)$ is

